

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2021

BIOLOGY PAPER 1

8:30 am – 11:00 am (2 hours 30 minutes)

This paper must be answered in English

GENERAL INSTRUCTIONS

- (1) There are **TWO** sections, A and B, in this Paper. You are advised to finish Section A in about 35 minutes.
- (2) Section A consists of multiple-choice questions in this question paper. Section B contains conventional questions printed separately in Question-Answer Book B.
- (3) Answers to Section A should be marked on the Multiple-choice Answer Sheet while answers to Section B should be written in the spaces provided in Question-Answer Book B. **The Answer Sheet for Section A and the Question-Answer Book B for Section B will be collected separately at the end of the examination.**

INSTRUCTIONS FOR SECTION A (MULTIPLE-CHOICE QUESTIONS)

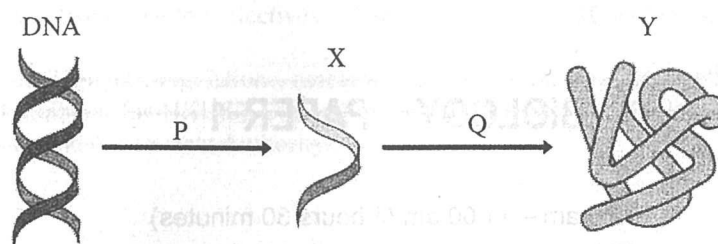
- (1) Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first stick a barcode label and insert the information required in the spaces provided. No extra time will be given for sticking on the barcode label after the 'Time is up' announcement.
- (2) When told to open this book, you should check that all the questions are there. Look for the words '**END OF SECTION A**' after the last question.
- (3) All questions carry equal marks.
- (4) **ANSWER ALL QUESTIONS.** You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answers clearly; otherwise you will lose marks if the answers cannot be captured.
- (5) You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.
- (6) No marks will be deducted for wrong answers.

Not to be taken away before the
end of the examination session

There are 36 questions in this section.

The diagrams in this section are NOT necessarily drawn to scale.

Directions: Questions 1 to 3 refer to the diagram below, which shows the flow of gene expression within a cell:



1. X represents

- A. a transfer RNA.
- B. a DNA template.
- C. a ribosomal RNA.
- D. a messenger RNA.

2. Y could be

- (1) an enzyme.
- (2) a hormone.
- (3) an antibody.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

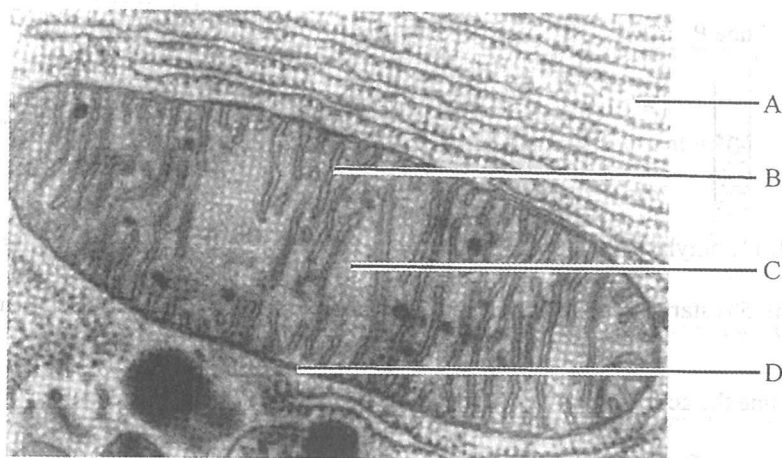
3. Which of the following combinations correctly states what process P is and where process Q takes place?

- | What process P is | Where process Q takes place |
|--------------------------|------------------------------------|
| A. translation | cytoplasm |
| B. translation | nucleus |
| C. transcription | cytoplasm |
| D. transcription | nucleus |

4. Which of the following pairs of molecules contains the greatest amount of stored energy?

- A. ADP and NAD
- B. ADP and NADH
- C. ATP and NAD
- D. ATP and NADH

Directions: Questions 5 and 6 refer to the electron micrograph below, which shows a mitochondrion in an animal cell:



5. Which of the following substances passes through D into the mitochondrion during aerobic respiration?

- A. glucose
- B. pyruvate
- C. acetyl-CoA
- D. triose phosphate

6. Anaerobic respiration takes place at

- A. A.
- B. B.
- C. C.
- D. D.

7. Which of the following correctly describe the absorption of amino acids in the small intestine?

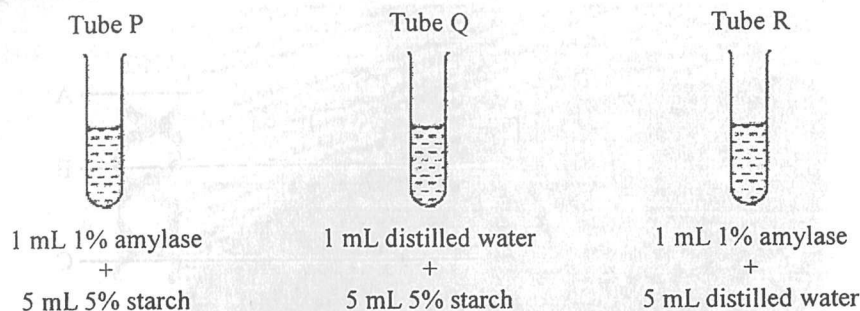
- (1) Amino acids are absorbed into lacteals.
- (2) The absorption is assisted by membrane proteins.
- (3) Amino acids can move along or against a concentration gradient.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

8. Which of the following combinations correctly lists the requirements of both osmosis and diffusion?

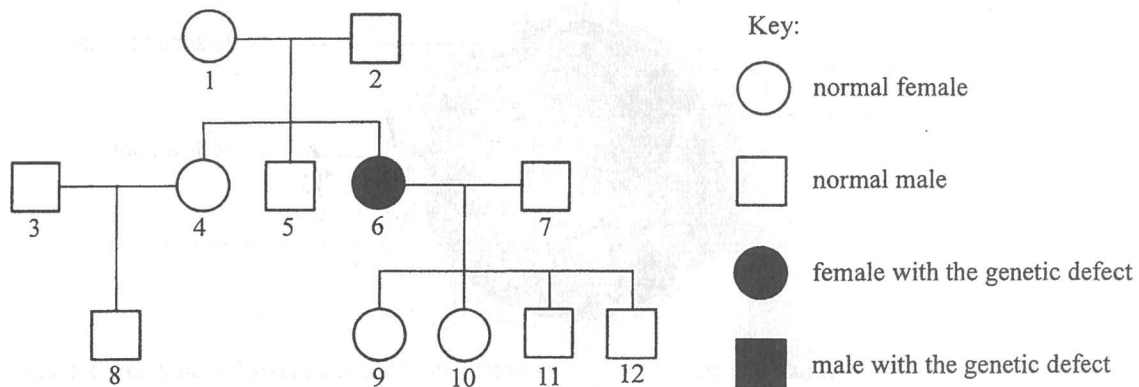
	<i>Concentration gradient</i>	<i>Energy supply</i>	<i>Selectively permeable membrane</i>
A.	Yes	No	Yes
B.	Yes	No	No
C.	No	Yes	Yes
D.	No	Yes	No

Directions: Questions 9 and 10 refer to the diagram below, which shows three set-ups used in the study of the activity of amylase:



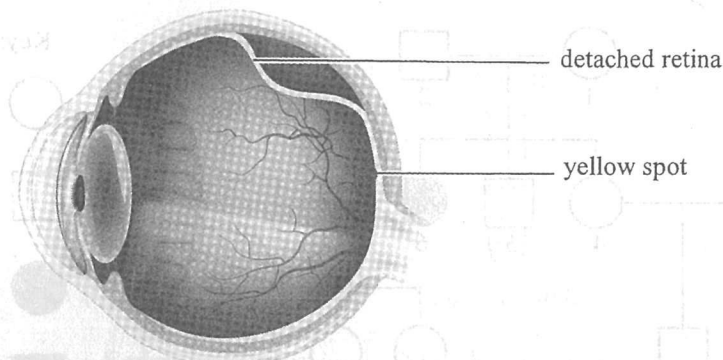
9. We can conclude the action of amylase on starch by comparing the results of tubes
- P and Q only.
 - P and R only.
 - Q and R only.
 - P, Q and R.
10. If the following tests are conducted on the reaction mixture of tube P, which test(s) will give the same result at the beginning of the experiment and after 10 minutes?
- (1) iodine test
 - (2) albutix paper
 - (3) Benedict's test
- (1) only
 - (2) only
 - (1) and (3) only
 - (2) and (3) only
11. The human ABO blood group system is controlled by multiple alleles. Alleles I^A and I^B lead to the presence of antigens A and B on the surface of red blood cells respectively. Allele i leads to the absence of antigens A and B on the surface of red blood cells. How many genotypes are involved in the ABO blood group system?
- 3
 - 4
 - 6
 - 8
12. A pair of identical twins were adopted by two different families. After the twins grow up, which of the following descriptions about them will most likely apply?
- They have different shapes of earlobe.
 - They have the same pattern of fingerprint.
 - They have different degrees of short sightedness.
 - They have the same number of freckles on their faces.

Directions: Questions 13 to 15 refer to the pedigree below, which shows the inheritance of a certain genetic defect controlled by a pair of alleles.

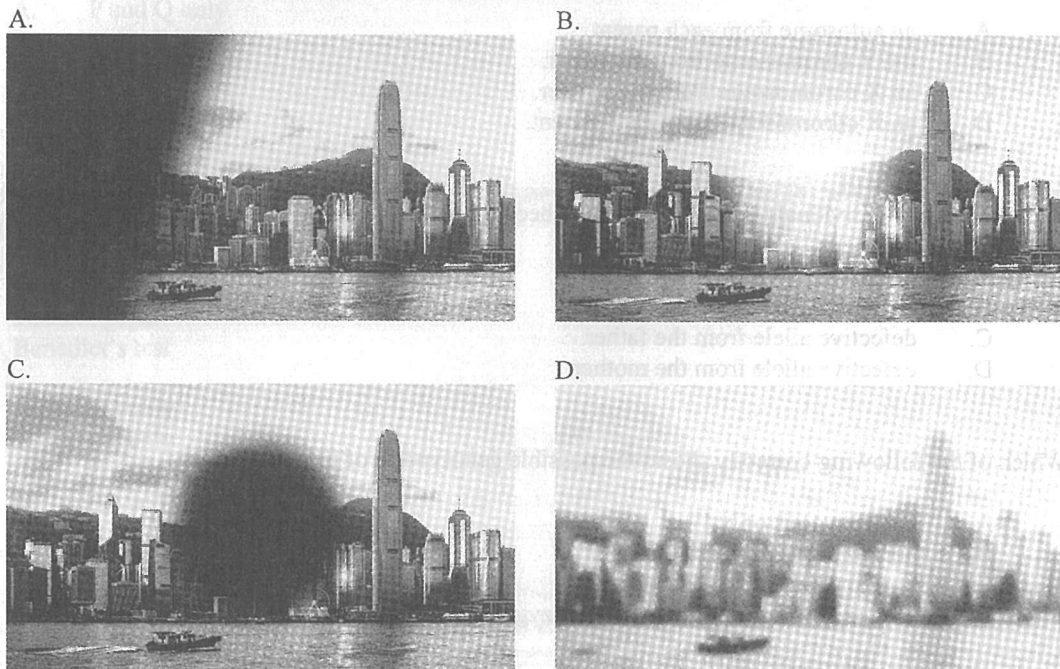


13. The phenotype of individual 6 can be explained by the fact that this individual received a defective allele on
 - A. an autosome from each parent.
 - B. the Y chromosome from the father.
 - C. an X chromosome from the mother.
 - D. an X chromosome from each parent.
14. The offspring of individuals 6 and 7 are normal because each receives a
 - A. normal allele from the father.
 - B. normal allele from each parent.
 - C. defective allele from the father.
 - D. defective allele from the mother.
15. Which of the following correctly shows the possible genotype(s) of individual 8?
 - A. heterozygous only
 - B. homozygous dominant only
 - C. homozygous recessive only
 - D. homozygous dominant and heterozygous
16. In the crossing of a red-flowered plant with a white-flowered plant, half of the F1 offspring were red-flowered and the other half were white-flowered. When the F1 white-flowered plants were self-crossed, all the F2 offspring were white-flowered. What can be concluded?
 - A. The red-flowered parental plant was homozygous.
 - B. The white-flowered parental plant was heterozygous.
 - C. White flower is the dominant character in this species of plant.
 - D. Red flower is the dominant character in this species of plant.
17. Although Mendel proposed that the characteristics of the pea plants were controlled by a pair of 'heredity factors', it was not recognised during his lifetime. His idea was later accepted in the early 20th century because it was supported by
 - A. the theory of natural selection.
 - B. the same patterns of inheritance in fruit flies.
 - C. the behaviours of chromosomes in cell division.
 - D. the discovery of the double helix structure of DNA.

18. The diagram below shows an early stage of an eye defect:

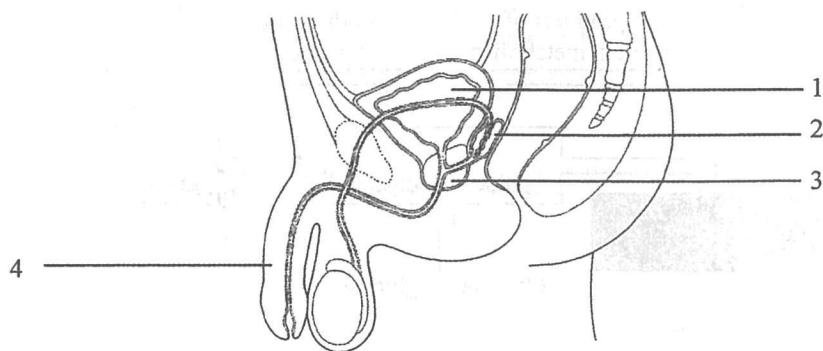


If a person suffers from this eye defect, which of the following diagrams is the most likely vision perceived by this person?



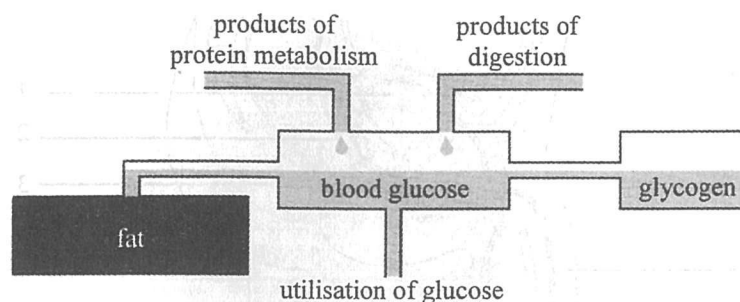
19. Which of the following segments of the alimentary canal absorbs the largest amount of water?
- the oesophagus
 - the stomach
 - the small intestine
 - the large intestine
20. Carbon dioxide produced in the small intestine is passed to the lungs for gas exchange. Which of the following correctly shows the transport route of the carbon dioxide?
- small intestine → lungs
 - small intestine → liver → lungs
 - small intestine → heart → lungs
 - small intestine → liver → heart → lungs

Directions: Questions 21 and 22 refer to the diagram below, which shows the male reproductive system and its associated structures:



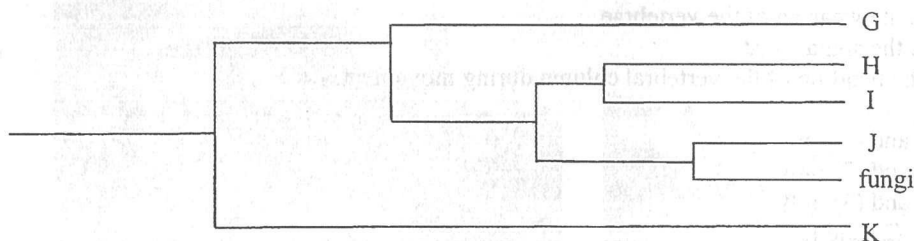
21. Which of the labelled structures are responsible for producing seminal fluid?
- 1 and 2 only
 - 1 and 3 only
 - 2 and 3 only
 - 1, 2 and 3
22. With reference to structure 4 above, which of the following structures of flowering plants has a similar function?
- style
 - anther
 - filament
 - pollen tube
23. Which of the following correctly describe the functions of the intervertebral disc?
- (1) It prevents the wearing of the vertebrae.
 - (2) It encloses the spinal cord.
 - (3) It allows the bending of the vertebral column during movement.
- (1) and (2) only
 - (1) and (3) only
 - (2) and (3) only
 - (1), (2) and (3)
24. Which of the following types of cells can be found in the tissue fluid?
- (1) B cell
 - (2) T cell
 - (3) phagocyte
- (1) and (2) only
 - (1) and (3) only
 - (2) and (3) only
 - (1), (2) and (3)
25. Which of the following is **not** the first line of defence against the invasion of pathogens in humans?
- saliva
 - sweat
 - mucus
 - lymph

Directions: Questions 26 and 27 refer to the model below, which shows the regulation of blood glucose level in humans:



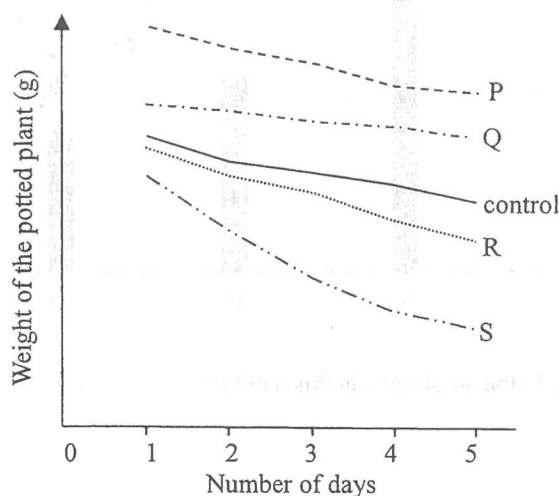
26. A student adds the following remarks to this model. Which remark is *incorrect*?
- Glycogen can be stored in muscles.
 - Insulin converts glucose to glycogen for storage.
 - The products of digestion take the form of simple sugars.
 - The products of protein metabolism come from the liver.
27. When the blood glucose level is higher than normal, which of the following changes is *incorrect*?
- The utilisation of glucose will increase.
 - The conversion of fat to glycogen will increase.
 - The conversion of blood glucose to fat will increase.
 - The conversion of blood glucose to glycogen will increase.

Directions: Questions 28 and 29 refer to the diagram below, which shows an evolutionary tree demonstrating the phylogenetic relationship of the six kingdoms:



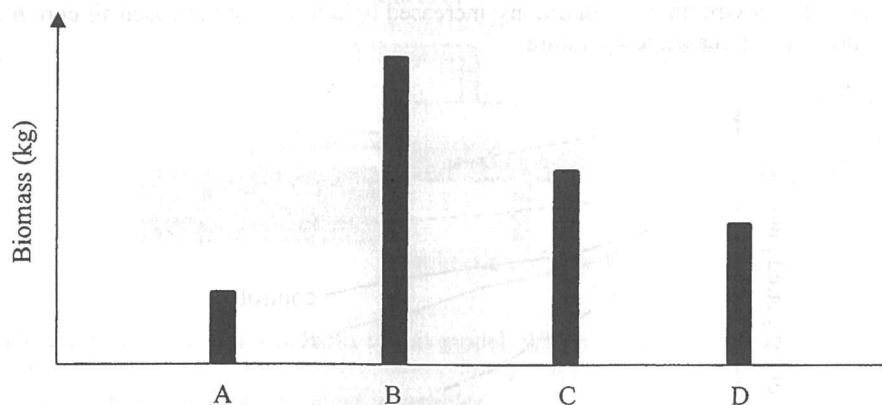
28. Which of the following combinations correctly shows the kingdoms represented by G, J and K in the evolutionary tree?
- | | G | J | K |
|----|-----------------|----------|------------|
| A. | Archaeobacteria | Animalia | Eubacteria |
| B. | Archaeobacteria | Plantae | Eubacteria |
| C. | Eubacteria | Plantae | Protista |
| D. | Eubacteria | Animalia | Protista |
29. Which of the following pairs of organisms belongs to the same domain?
- G and K
 - G and H
 - H and J
 - J and K

Directions: Questions 30 and 31 refer to the graph below, which shows the effect of environmental conditions on the transpiration rate of a potted plant placed in a small room. The weight of the potted plant was recorded daily at noon for five consecutive days (control experiment). The experiment was repeated by changing one of the environmental conditions: increased light intensity, increased air current, increased relative humidity, or increased temperature.



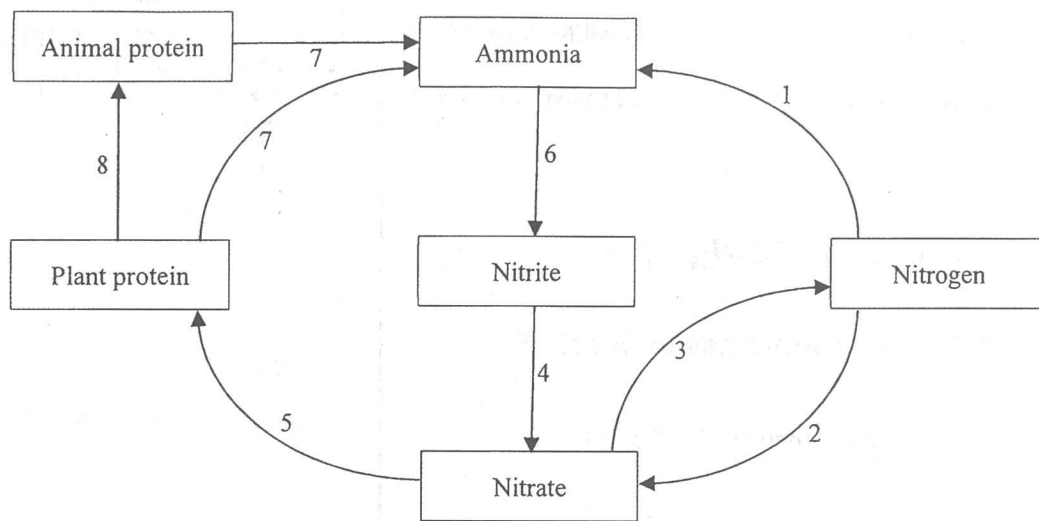
30. Which of the following lines represents the results with an increased relative humidity?
- P
 - Q
 - R
 - S
31. In the above study, which of the following steps is necessary?
- Remove any fallen leaves.
 - Wrap the pot with a plastic bag.
 - Water the plant every day in the morning.
 - Smear vaseline on the lower epidermis of the leaves.
32. During primary succession in a terrestrial habitat, which of the following descriptions about the ecosystem is *incorrect*?
- The nutrient level of the soil decreases.
 - The complexity of the food web increases.
 - The abundance of non-vascular plants decreases.
 - The energy available to the community increases.

Directions: Questions 33 and 34 refer to the following graph, which shows the biomasses of four populations forming a food chain in a terrestrial habitat:



33. Which population is most likely the autotrophs in this habitat?
- A. A
 - B. B
 - C. C
 - D. D
34. What would you expect to happen if the number of individuals in population D decreased?
- A. The number of individuals in population A would increase.
 - B. The number of individuals in population C would decrease.
 - C. The amount of energy passed to population A would decrease.
 - D. The amount of energy passed to population C would increase.

Directions: Questions 35 and 36 refers to the diagram below, which shows some natural processes involved in the nitrogen cycle:



35. Process 2 refers to

- A. lightning.
- B. nitrification.
- C. denitrification.
- D. nitrogen fixation.

36. Which of the following pairs of processes involves the action of bacteria?

- A. 1 and 5
- B. 2 and 6
- C. 3 and 7
- D. 4 and 8

END OF SECTION A

Go on to Question-Answer Book B for questions on Section B